

Demonstrating Mid-Air Ultrasound Haptics with Thermal Display

Yatharth Singhal*

The University of Texas at Dallas

Haokun Wang†

The University of Texas at Dallas

Jin Ryong Kim‡

The University of Texas at Dallas

ABSTRACT

We demonstrate an ultrasound haptic-based mid-air thermo-tactile display system. We design a proof-of-concept thermo-tactile feedback system with an open-top chamber, heat modules, and an ultrasound haptic display. Our method involves directing heated airflow toward the focused pressure point produced by the ultrasound display to deliver thermal and tactile cues in mid-air simultaneously. We demonstrate our system with four different VR environments (*campfire*, *water fountain*, *kitchen*, and *candle*) to show the rich user experiences of integrating thermal and tactile feedback.

1 INTRODUCTION

Virtual Reality (VR) is experiencing a surge in popularity and adoption, with increasing demand across entertainment, education, and industry sectors, signifying its upward trajectory and expanding influence. Integrating thermal feedback plays a pivotal role in enhancing the overall immersive experience in VR. Thermal feedback adds an extra dimension to the sensory realism, providing users with a more comprehensive and immersive environment. In VR applications, integrating thermal feedback can simulate the sensation of hot or cold, creating a more realistic and engaging experience, especially in scenarios where temperature perception is crucial for immersion, such as exploring different environments or interacting with virtual objects that emit heat.

Various research initiatives have focused on implementing technologies like Peltiers, infrared lasers, thermal radiation, liquid systems, and mid-air haptic displays to provide users with thermal feedback on various body parts [1, 4, 5]. Despite significant advancements in thermal feedback system development, numerous challenges persist. These include issues like bulky systems with large actuators and intricate designs, which impede a natural user experience, along with the high costs and substantial power consumption associated with thermal devices [1]. Another challenge is the limited interaction space due to the small thermal actuator size, resulting in a restricted scale of thermal display [4]. Motoyama et al. [3] used mist to provide cold sensations on the fingertip. Kamigaki et al. [2] proposed using an ultrasound display to create a focal point in mid-air on a hand-worn glove to provide thermal sensations. However, due to their underlying mechanisms, most of the above-mentioned approaches have the limitation that it is difficult to control the thermal cues in 3D space. Other approaches that provide controllability do not offer bare-hand interaction.

Our perception-based approach integrates localized ultrasound tactile feedback alongside global thermal feedback to generate a combined thermo-tactile sensation experienced directly on bare hands within a mid-air environment. The tactile stimulus is detected through mechanoreceptors and processed in the somatosensory cortex, while the thermal stimulus is perceived via thermoreceptors and processed in the insula cortex. By harmonizing these two stimuli in a manner that doesn't mask each other while targeting the same

*e-mail: Yatharth.Singhal@UTDallas.edu

†e-mail: Haokun.Wang@UTDallas.edu

‡e-mail: Jin.Kim@UTDallas.edu

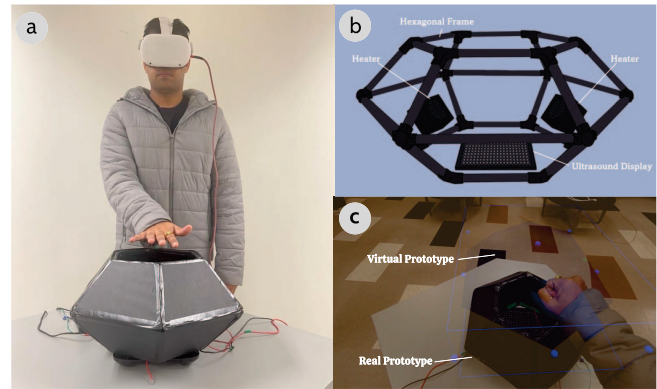


Figure 1: (a) A user interacting with the actual system, (b) a diagram of the proposed system, and (c) a snapshot of the calibration process

area of the human hands, users can perceive a unified thermo-tactile feedback, enabling the provision of thermo-tactile cues in three-dimensional space at any chosen location. Our method effectively merges vibrotactile and thermal stimuli without mutual interference, achieved through the creation of ambient thermal cues using a heated chamber. By combining acoustic pressure points with ambient thermal cues, we attain integrated thermo-tactile sensations in mid-air for barehand interaction. For further details about our system and the experiments validating our approach, refer to [5, 6].

2 SYSTEM

2.1 Hardware

Figure 1 shows the design of our proof-of-concept prototype. It consists of two heat modules, an ultrasound display, and an open-top chamber for retaining heat inside while allowing heat to flow upward. The system delivers thermal stimulus on the user's hand using a PTC (Positive Thermal Coefficient) heating element to proportionately raise its resistance to maintain a constant temperature. The temperature of the heater is controlled by varying the input voltage of a power supply. The chamber boasts a hexagonal shape, deliberately designed to facilitate thermal flow. Its construction incorporates 30 aluminum beams forming the frame of the structure, while the surface cover utilizes insulation foams for effective insulation. The top is cut out in the shape of a regular hexagon of side 300 mm and an ultrasonic haptic display (STRATOS Explore, Ultraleap, UK) is placed inside the chamber, facing upward. Oculus hand tracking is used to precisely track users' hands and deliver thermo-tactile feedback. When the ultrasound haptic display is activated, the focused ultrasonic haptic cues with different temperatures are created through the open top of the chamber, passing through the heated air. Participants can interact with an apparatus by horizontally outstretching their hands in an open space between the apparatus's top surface and roughly 100 mm above it, which we refer to as *interaction space*. In addition to this, a Meta Quest 3 VR headset is seamlessly integrated with our setup to provide a comprehensive VR experience enriched with thermo-tactile feedback.



Figure 2: User interaction with thermal objects: (a) campfire, (b) water fountain, (c) steam, and (d) candle.

2.2 VR Scenes

We demonstrate our approach using four VR scenes, each providing unique thermo-tactile feedback. To provide an exemplary user experience, we perform a calibration process aiming to precisely align the prototype with the user's position within the virtual scenes. We utilize Mixed Reality passthrough capabilities to enable the simultaneous visualization of both physical and virtual prototypes, overlaying them to align and calibrate the users' position and location. We describe each environment in detail below:

Campfire: Figure 2 (a) shows the campfire scene. In this scene, the user interacts with the fire embers and smoke rising from the fire with their virtual hands. An ultrasonic cue with 200 Hz and 0.7 intensity for haptic rendering is provided for 500 ms whenever an ember collides with the virtual hands. The temperature for this scene is set to 44°C to maximize the heat effect without reaching the pain threshold to resemble the fire. A crackling fire sound is played with the virtual scene to enhance the user experience.

Water Fountain: In this scene, a hot water fountain with a water stream is presented in a park (see Figure 2 (b)). When the water stream collides with the user's hands, the height of the water stream is dynamically adjusted depending on the hand's position with the water splash. A continuous circular haptic sensation with a radius of 2 cm with a frequency of 200 Hz and an intensity of 1.0 is presented to the user's palm. A temperature of 40°C is selected to emulate the feel of warm water hitting the palm.

Steam: The users find themselves in a kitchen and see a teapot on the stove with steam coming out of it (see Figure 2 (c)). The participants can interact with the steam by using their virtual hands. A haptic cue of radius 4 cm with a frequency of 200 Hz and intensity of 0.4 is presented on the user's palm to provide haptic sensations of steam interacting with the palm. The temperature for this scene is set to 38°C to provide the heat effect that resembles the steam.

Candle: In this scene, a candle with a bright flame is placed on a desk (see Figure 2 (d)). The participants can put their hands above the candle flame to perceive the fluctuating hot air. A circular haptic cue sensation with a radius change from 2 cm to 0.5 cm in 2 seconds with a frequency of 200 Hz and intensity change from 1.0 to 0.8 is provided at the height of 25 cm above the ultrasonic board. A temperature of 36°C is selected to simulate the flicking hot air tapping the palm.

3 DEMO EXPERIENCE

During the conference, our demonstration will feature the proposed thermo-tactile display within a VR setup. Interested attendees will receive guidance on interacting with the system, involving standing in front of it and donning the headset. Demonstrators will then facilitate a calibration process to synchronize the attendee's physical and virtual positions. Attendees can subsequently engage with the prototype, exploring and interacting with various objects within the virtual environment. The entire demonstration, including calibration and briefing, is expected to last no more than 3 minutes per attendee. This approach ensures an engaging and efficient interaction with our thermo-tactile VR system.

ACKNOWLEDGEMENTS

This work was supported by Electronics and Telecommunications Research Institute (ETRI) grant funded by the Korean government [24ZC1200, Research on hyper-realistic interaction technology for five senses and emotional experience].

REFERENCES

- [1] S. Günther, F. Müller, D. Schön, O. Elmoghazy, M. Mühlhäuser, and M. Schmitz. Terminator: Understanding the interdependency of visual and on-body thermal feedback in virtual reality. In *Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems*, pp. 1–14, 2020. 1
- [2] T. Kamigaki, S. Suzuki, and H. Shinoda. Mid-air thermal display via high-intensity ultrasound. In *SIGGRAPH Asia 2020 Emerging Technologies*, pp. 1–2. 2020. 1
- [3] H. Motoyama, M. Fujiwara, T. Morisaki, H. Shinoda, and Y. Makino. Touchable cooled graphics: Midair 3d image with noncontact cooling feedback using ultrasound-driven mist vaporization. In *SIGGRAPH Asia 2022 Emerging Technologies*, pp. 1–2. 2022. 1
- [4] R. L. Peiris, W. Peng, Z. Chen, L. Chan, and K. Minamizawa. Thermovr: Exploring integrated thermal haptic feedback with head mounted displays. In *Proceedings of the 2017 CHI Conference on Human Factors in Computing Systems*, pp. 5452–5456, 2017. 1
- [5] Y. Singhal, H. Wang, H. Gil, and J. R. Kim. Mid-air thermo-tactile feedback using ultrasound haptic display. In *Proceedings of the 27th ACM Symposium on Virtual Reality Software and Technology*, pp. 1–11, 2021. 1
- [6] Y. Singhal, H. Wang, and J. R. Kim. Fire: Mid-air thermo-tactile display. In *SIGGRAPH Asia 2023 Emerging Technologies*, pp. 1–2. 2023. 1